

AB POWER ENGINEERING

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INTRODUCTION

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1. Focuses on the application of power and energy sources to enhance agricultural practices. a. Agricultural Power Engineering b. Agricultural machinery Engineering c. Agricultural Mechanization d. All of the above 2. Agricultural Power and Energy Engineering focuses on applying engineering principles to enhance agricultural productivity and sustainability by optimizing energy use and developing efficient power systems a. Agricultural Power and Energy Engineering b. Agricultural machinery Engineering c. Agricultural Mechanization d. All of the above 3. Coverage of study of power engineering. a. Design, development, and management of machinery, equipment, and systems. b. Addresses issues related to energy efficiency, resource conservation, and environmental sustainability in agriculture c. All of the above	4. Key aspect of agricultural power engineering a. Designing and optimizing farm equipment which involves understanding machinery efficiency, energy sources (human, animal, mechanical, renewable), and mechanization levels. b. Developing and managing efficient water systems for crop production. c. Exploring and implementing renewable energy sources like solar, wind, and biomass for on-farm energy needs, reducing reliance on fossil fuels. d. Designing farm buildings, storage facilities, and greenhouses, as well as developing systems for biogas generation and waste management. e. Optimizing processes and machinery for handling, storage, processing, and packaging of agricultural products. f. Mitigating the environmental impact of farming practices through sustainable technologies and waste management solutions g. Implementing automation and control systems in agricultural operations to improve efficiency and productivity. h. All of the above
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5. Specific subjects in agricultural power engineering. a. Farm machinery and equipment b. Agricultural power units c. Fluid mechanics d. Soil and water conservation engineering e. Agricultural processing and storage f. Renewable energy in agriculture g. Agricultural structures and environmental science h. Instrumentation and control i. Agricultural systems engineering j. All of the above 6. Importance of power in agriculture. a. Increase productivity b. Reduce labor cost c. Timely operation d. Improved quality e. Increased income f. All of the above	7. Power sources in agriculture a. Mobile power b. Stationary power c. Renewable Energy d. All of the above 8. What are the mobile power in agriculture a. Human labor b. Draft animals c. Tractors d. Self propelled machinery e. All of the above 9. What are the stationary power used in agriculture? a. Oil engine (gasoline, diesel, and kerosene) b. Electric motors c. All of the above
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10. Power sources in agriculture.

- a. Human
- b. Animal power
- c. Mechanical power
- d. Electrical power
- e. Renewable energy
- f. All of the above

Power = 0.1 hp
Female = 0.075 hp

11. Which of the following can be adopted for efficient and sustainable power farming operation?

- a. Optimizing power usage
- b. Apply precision farming
- c. Integrate renewable energy
- d. All of the above

12. Commonly use energy sources in agriculture.

- a. Fossil fuel
- b. Electricity
- c. Biomass
- d. All of the above

13. The act regulating the practice of agricultural engineering in the Philippines.

- a. Republic Act No. 8559. (New - RA10915)
- b. Republic Act No. 9513
- c. Republic Act No. 8749
- d. None of the above

14. A facility that produces electrical and/or mechanical energy and forms of useful thermal energy such as heat and steam, which are used for heating and cooling purposes.

- a. Co-firing system
- b. Co-generation system
- c. Hybrid generation system
- d. None of the above

15. A facility for the production of electricity and/or thermal energy.

- a. Renewable energy facility
- b. Generation facility
- c. Small power facility
- d. None of the above

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16. The term referred to any power or energy generation facility that makes use of two or more types of technologies utilizing both conventional and renewable fuel sources.

- a. Co-generation system
- b. Hybrid system
- c. Geothermal system
- d. None of the above

17. Small generation entities directly supplying electricity to distribution grid, not exceeding 100 kW in capacity.

- a. Co-generation system
- b. Distributed generation system
- c. Biomass generation system
- d. None of the above

18. Which of the following is a non-renewable fuel?

- a. Crude oil
- b. Natural gas - 75
- c. Coal 5
- d. All of the above
- e. None of the above

19. An energy source that is commonly used, like crude oil and geothermal.

- a. Renewable energy
- b. Non-renewable energy
- c. Conventional energy
- d. Non-conventional energy
- e. None of the above

20. Which of the following comes from crude oil?

- a. Gasoline
- b. Kerosene
- c. Diesel
- d. All of the above

21. A device that converts the kinetic energy of steam into mechanical power.

- a. Boiler
- b. Steamer
- c. Turbine
- d. None of the above

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22. A device that converts heat into steam providing kinetic energy to produce mechanical power.

- a. Steamer
- ☒ b. Boiler
- c. Heater
- d. None of the above

23. A device that converts heat into electricity.

- a. Solar cell
- b. Solar collector
- ☒ c. Peltier chip
- d. None of the above

24. A mass of molecules that produces power either heat, mechanical, electrical, chemical or any of their combination.

- a. Wood
- b. Gasoline
- c. Diesel
- d. LPG
- ☒ e. All of the above
- f. None of the above

25. Goals of agricultural power.

- a. Increase Production and Food Security
- b. Alleviate poverty and Improve Livelihood
- c. Generate Employment
- d. Reduce Drudgery
- ☒ e. All of the above

26. Liquefied natural gas (LNG) has ____ heat content than biogas and producer gas.

- ☒ a. lower
- b. higher
- c. the same
- d. None of the above

27. Which of the following fuel contains methane and carbon dioxide gases?

- a. Producer gas
- b. Synthetic gas
- ☒ c. Natural gas
- d. None of the above

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28. A device used to convert mechanical power to electrical power.

- a. Motor
- ☒ b. Generator
- c. Engine
- d. None of the above

29. A device used to convert electrical power to mechanical power.

- ☒ a. Motor
- b. Generator
- c. Alternator
- d. None of the above

30. A device that converts the potential energy of water into mechanical power that drives a generator.

- ☒ a. Turbine
- b. Pump
- c. Spinner
- d. None of the above

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GOD BLESS!!

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